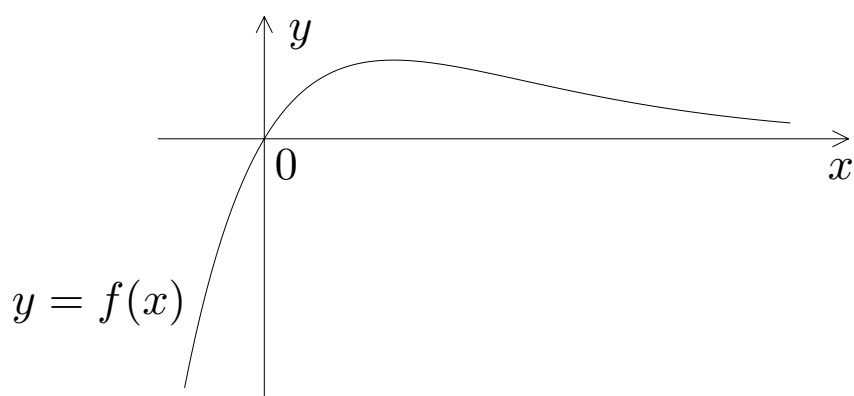


MATH1081 DISCRETE MATHEMATICS

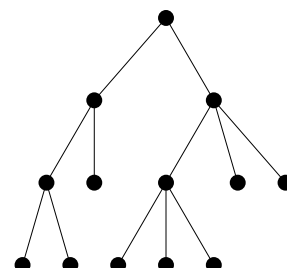
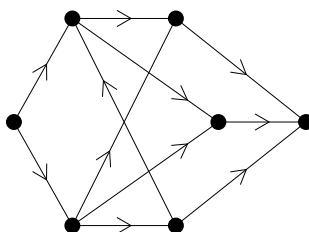
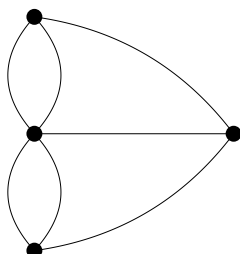
Section 5

Graphs

For this section of the course a graph is not something like this,



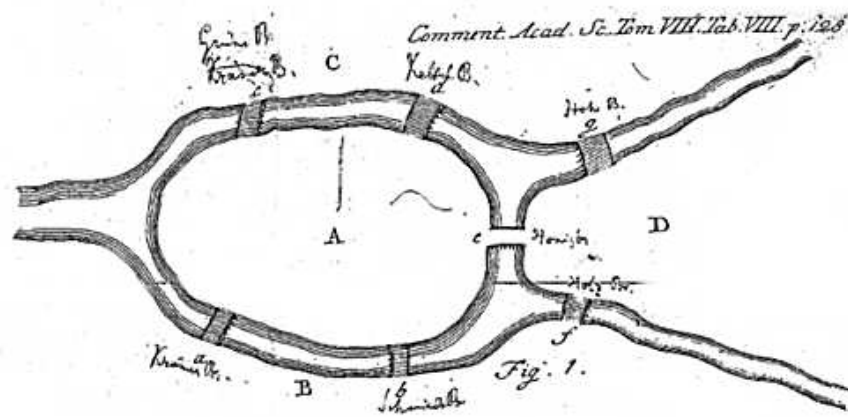
but like these.



Roughly, a **graph** consists of any finite number of points called **vertices** (singular **vertex**), some pairs of which may be joined by lines called **edges**.

Some uses of graphs.

- Transport problems:
 - the Königsberg bridges problem;



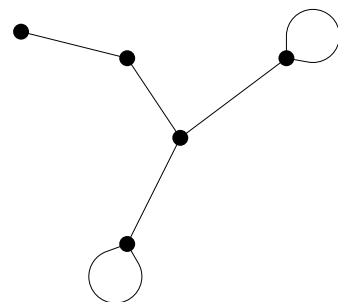
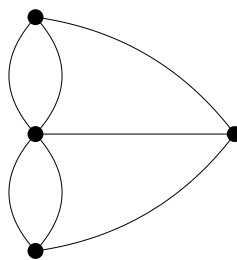
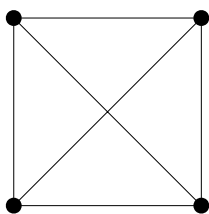
- the travelling salesman problem;
- minimal cost electricity grid.
- Activity analysis.
- Theory of automata.
- Matching and scheduling problems.
- Chemistry.
- Electrical circuits.
- Probability and counting.
- Practice at proofs!

Definitions.

- A **graph** G consists of two finite sets V and E . The elements of V are called the **vertices** of the graph, and the elements of E , the **edges**. Each edge is associated with either one or two vertices called its **endpoints**.
- An edge is called a **loop** if it has only one endpoint.
- Two (or more) edges with the same endpoints as each other are called **parallel** or **multiple** edges.
- A **simple** graph is one with no loops or parallel edges.

Comment. To state more formally the idea that “each edge is associated with either one or two vertices”, we may define a graph to consist of the sets V and E , together with a function $f : E \rightarrow \mathcal{P}(V)$ such that for every e , the set $f(e)$ has either one or two elements.

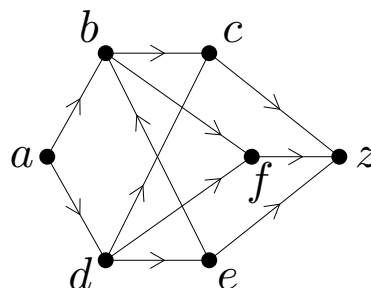
Examples.



Sometimes it is necessary to consider **directed graphs**, in which each edge has a direction associated with it.

Examples.

- An interesting example of a directed graph is the **internet graph** or **webgraph**, which has a vertex representing every page on the internet (at some specific time), and an edge from u to v whenever page u has a link to page v . There is a bit of information on Wikipedia, and plenty of images of (part of) this graph.
- Here is a more “arbitrary” example. Graphs like this can be used in planning and scheduling problems.

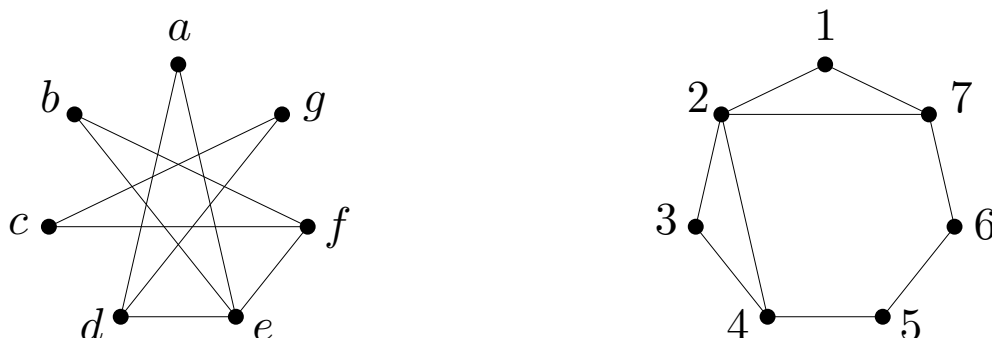


- A Hasse diagram is a directed graph in which the direction of an edge indicates which element is related to which. Although the edges are not customarily marked with arrows, we always draw the diagram in such a way that the edges are directed upwards.

Basically, the only relevant features of a graph are the number of its vertices and the way in which they are joined by edges. The following diagrams represent the same graph.



Likewise, the following graphs can be seen to be the same if they are labelled appropriately.

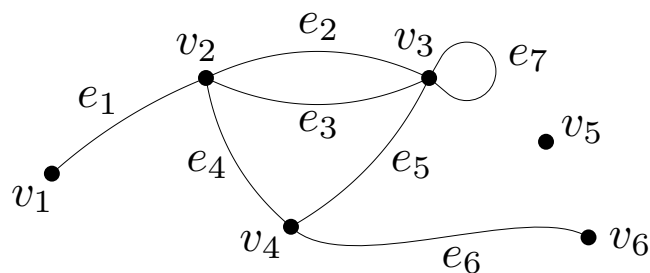


If the diagrams were labelled differently they might not, strictly speaking, represent the same graph. However they would have the same number of vertices, connected in the same manner, and indeed the only difference would be in the names of the vertices and edges. Such graphs are called **isomorphic**, and for most purposes can be regarded as “the same”.

More terminology.

- Two vertices joined by an edge are called **adjacent**.
- An edge is said to be **incident** on each of its endpoints.
- A vertex without incident edges is **isolated**.
- The **degree**, denoted $\deg(v)$, of a vertex v is the number of edges incident on v , with loops counting twice.

Example. Consider the following graph G .



- G contains a loop, e_7 , and parallel edges e_2 and e_3 ; thus G is not a simple graph.
- e_5 is incident on v_3 and v_4 .
- v_2 is adjacent to v_3 ; also, v_3 is adjacent to itself.
- v_5 is an isolated vertex.
- $\deg(v_4) = 3$ and $\deg(v_3) = 5$.

Exercise. Can you draw graphs with five vertices, having the following degrees?

(a) 4, 2, 2, 1, 1;

(b) 4, 4, 3, 2, 1;

(c) 4, 3, 3, 2, 1.

Theorem. *The Handshaking Lemma.* In any graph, the sum of the degrees of all the vertices equals twice the number of edges.

Proof. For every edge, count both its endpoints (the same one twice in the case of a loop). Since each edge has two endpoints, the total is twice the number of edges. But every vertex v has been counted once for each edge incident on it (twice for each loop). That is, v has been counted $\deg(v)$ times, and so the total is the sum of all the degrees. Hence the sum of all degrees in the graph equals twice the number of edges.

Exercise. Give an alternative proof of this result, using mathematical induction on the number of edges.

Corollary. The sum of the degrees in any graph is even.

Theorem. In any graph, the number of vertices having odd degree is even.

Proof by contradiction. Suppose that G is a graph in which the number of vertices having odd degree is odd. Then

- the sum of the degrees of these vertices is an odd number of odd numbers, which is odd;
- the sum of the degrees of the remaining vertices is the sum of a collection of even numbers, which is even;

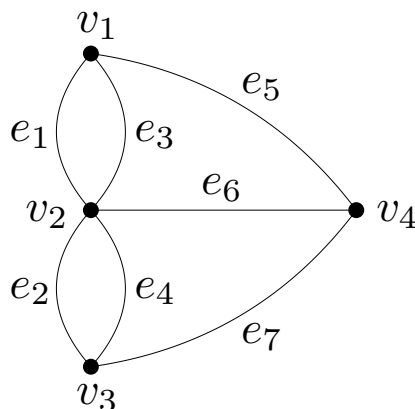
so the sum of all degrees in G is an odd number plus an even number, which is odd. But this has just been proved impossible. Therefore our original assumption is wrong, and the number of vertices having odd degree must be even.

Given a graph G , a subgraph of G is just a graph which is “part of” G . More formally,

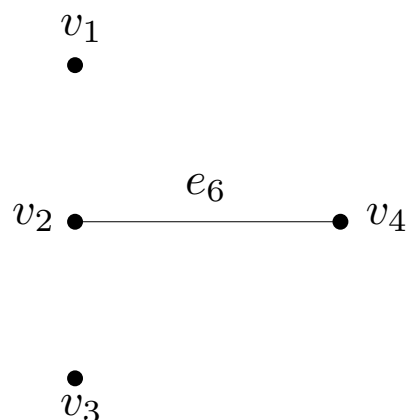
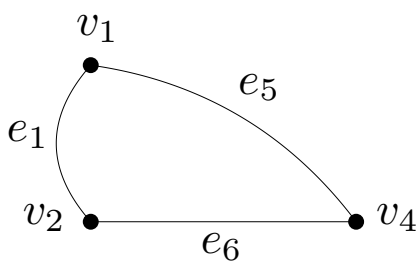
Definition. Let G be a graph with vertices V and edges E . A subgraph of G is a graph H with vertices V' and edges E' , such that

- $V' \subseteq V$;
- $E' \subseteq E$;
- for any e in E' , if e is incident on v in G , then e is incident on v in H .

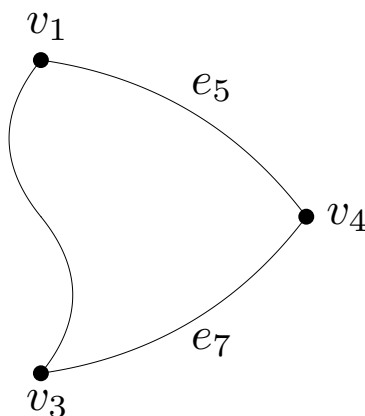
Example. The graph G ,



has many subgraphs, including these:



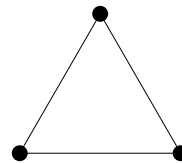
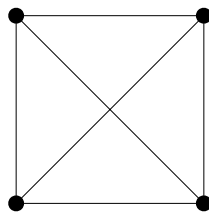
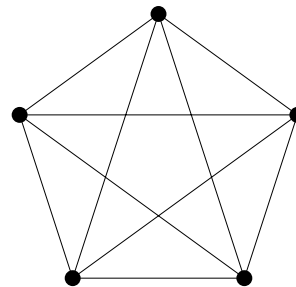
However the following is not a subgraph of G .



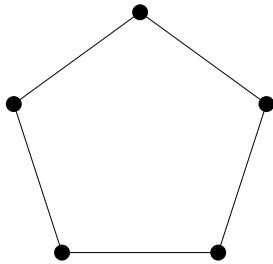
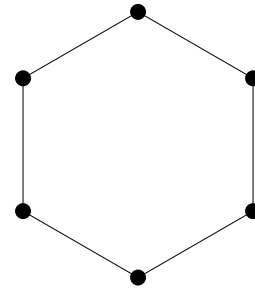
Problem. How many different (labelled) subgraphs does G have?

Special graphs.

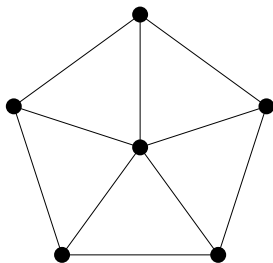
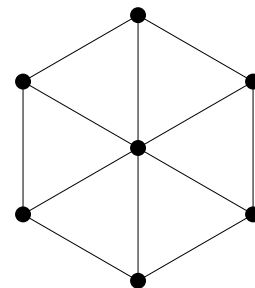
The **complete graph** on n vertices, denoted K_n , is the graph consisting of n vertices, with exactly one edge joining each pair of distinct vertices. The complete graphs K_2 , K_3 , K_4 and K_5 are shown below.

 K_2  K_3  K_4  K_5

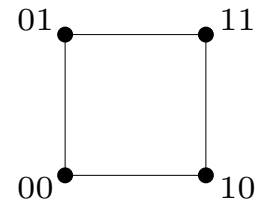
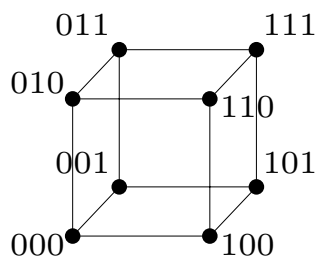
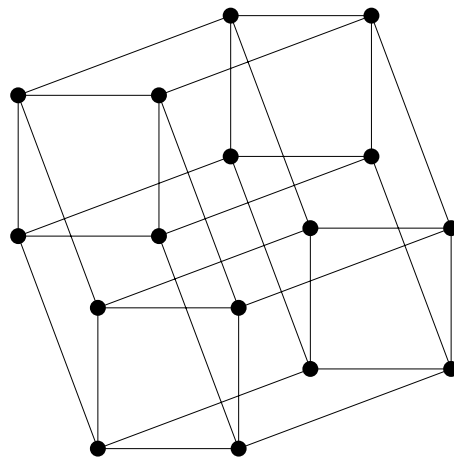
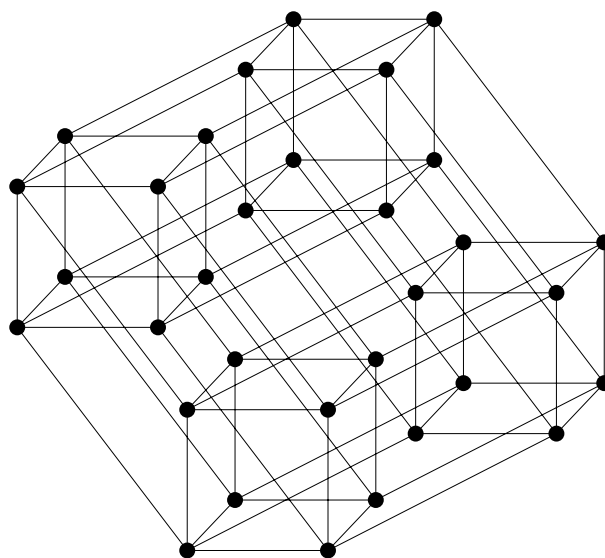
The **cycle** C_n , for $n \geq 3$, consists of vertices $v_1, v_2, \dots, v_n$ and edges $v_1v_2, v_2v_3, \dots, v_{n-1}v_n$ and v_nv_1 .

 C_5  C_6

The **wheel** W_n , for $n \geq 3$, consists of the cycle C_n together with another vertex v_0 adjacent to each of the vertices $v_1, v_2, \dots, v_n$ of the cycle.

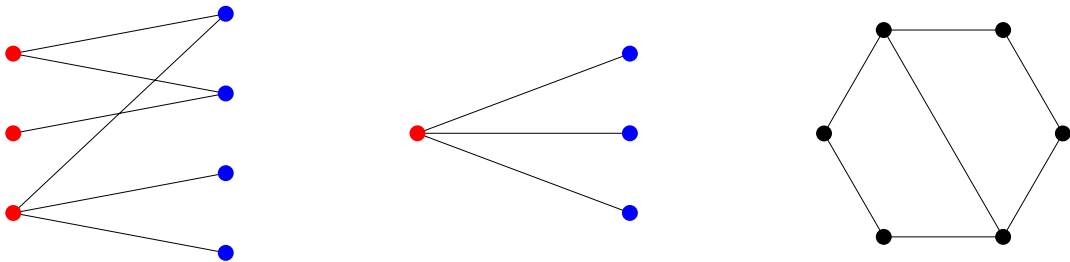
 W_5  W_6

The n -**cube** Q_n has 2^n vertices labelled with the 2^n bit strings of length n . Two vertices are adjacent if and only if their labels differ in exactly one place.

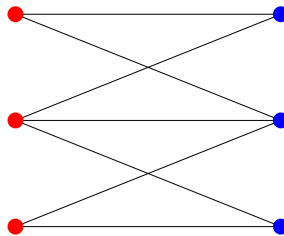
 Q_1  Q_2  Q_3  Q_4  Q_5

Definition. A **bipartite** graph is a simple graph in which the vertices can be partitioned into two (disjoint, non-empty) sets in such a manner that no two vertices in the same set are adjacent.

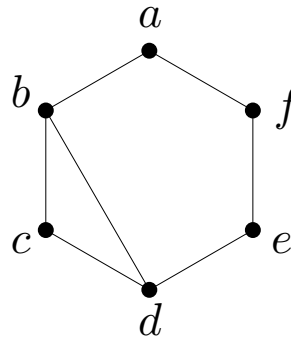
The following graphs are bipartite.



The third graph is seen to be bipartite on redrawing it isomorphically in **standard form** as follows.



Example. Prove that the following graph is not bipartite.



Solution. Suppose that the graph is bipartite. Partition the vertices into two sets in such a way that no two vertices in the same set are adjacent. Now b is in one of these sets; but b is adjacent to c and d , so both of these must be in the other set. But this is impossible since c and d are adjacent to each other. This contradiction proves that the graph is not bipartite.

Exercise.

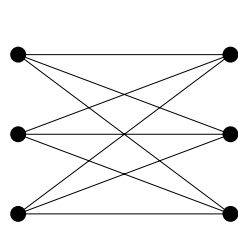
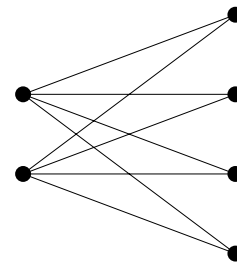
- (i) For which values of $n \geq 3$ is the cycle C_n bipartite?
- (ii) For which values of $n \geq 3$ is the wheel W_n bipartite?
- (iii) For which values of $n \geq 1$ is the n -cube Q_n bipartite?

Answers.

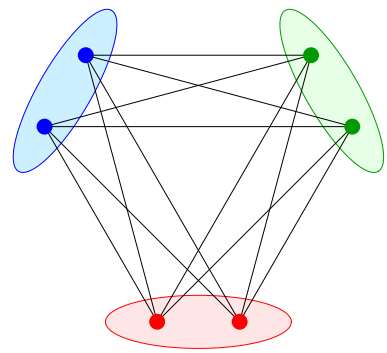
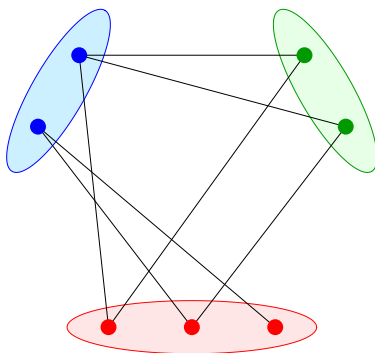
- (i) For $n \geq 3$, the cycle C_n is bipartite if and only if n is even.
- (ii) For $n \geq 3$, the wheel W_n is never bipartite.
- (iii) For $n \geq 1$, the n -cube Q_n is always bipartite.

Proof of (iii). Let O be the set of all vertices whose labels contain an odd number of zeros and E the set of all vertices whose labels contain an even number of zeros. Then O and E are disjoint and $O \cup E$ is the set of all vertices of Q_n . Let u and v be vertices in O with $u \neq v$. Then u and v have labels differing in at least two places, and therefore are not adjacent in Q_n . Similarly, no two vertices in E are adjacent. Thus Q_n is bipartite.

Definition. The **complete bipartite graph** $K_{m,n}$ is the simple graph whose vertices can be partitioned into two sets containing m and n elements in such a way that two vertices are adjacent if and only if one is in the first set and the other in the second.

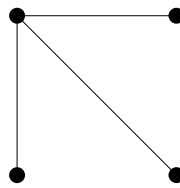
 $K_{3,3}$  $K_{2,4}$

We may also speak of **tripartite** graphs and **complete tripartite graphs**.

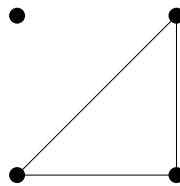
 $K_{2,2,2}$

Definition. Given any simple graph G , its **complement** (or *complementary graph*) is the simple graph $\overline{G}$ which has the same vertices as G , and in which two vertices are adjacent if and only if they are *not* adjacent in G .

Example. The complement of



is



Exercise. A football competition includes six teams. Prove that at any time in the season there must be *either* a set of three teams which have all played each other, or a set of three teams in which no two have played each other.

Solution. Draw a graph G in which the vertices are labelled with the names of teams, and two vertices are adjacent if and only if the teams have played each other. Then in the complement $\overline{G}$ two vertices are adjacent if and only if the teams have not played each other.

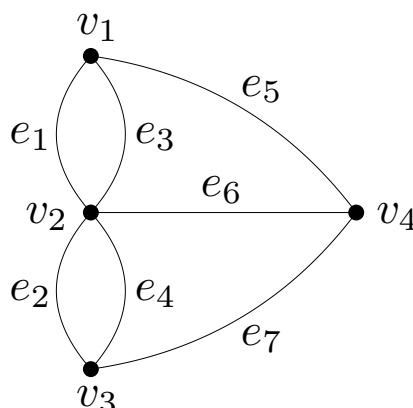
Choose any vertex v .

Case 1, v is adjacent to three (or more) vertices w_1, w_2, w_3 in G . If two of w_1, w_2, w_3 are adjacent then these two, together with v , form a set of three teams which have all played each other. Otherwise w_1, w_2, w_3 constitute a set of three, no two of whom have played each other.

Case 2, v is adjacent to two or fewer vertices in G . Now G contains five vertices other than v , and so v must be adjacent to three or more vertices in $\overline{G}$. Therefore we can apply the same argument as before.

PATHS AND CIRCUITS

The Königsberg bridges can be represented thus:



A possible walk around the town could be written

$$v_1, e_5, v_4, e_6, v_2, e_2, v_3, e_2, v_2.$$

Definition.

- A **walk** from u to v in a graph G is a finite sequence of vertices and edges alternately,

$$v_0, e_1, v_1, e_2, v_2, \dots, v_{n-1}, e_n, v_n,$$

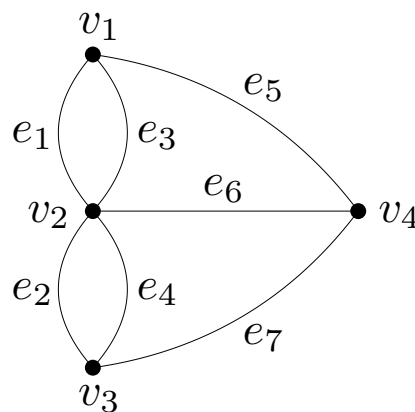
where each edge e_i is incident on vertices v_{i-1} and v_i , and $v_0 = u$ and $v_n = v$. The **length** of this walk is n .

- A **path** is a walk in which all the edges are different.
- A **circuit** is a path which begins and ends at the same vertex.
- A path is **simple** if it has no repeated vertices. A circuit is simple if it has no repeated vertices *except* that the first vertex is the same as the last.

Notes.

- The above terminology is not completely standardised. If you are reading any alternative sources you will need to make a careful note of their terminology.
- In many cases, a walk can be unambiguously specified by listing its edges alone; and in a simple graph, a walk can be specified by listing its vertices.

Examples. In the Königsberg bridges graph,

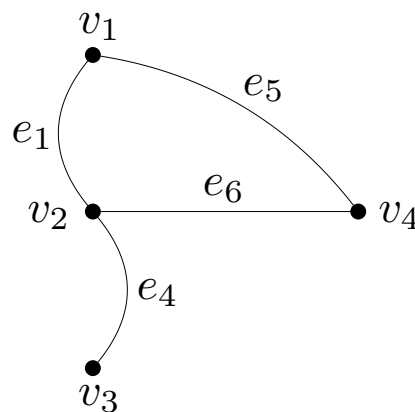


- $v_1, e_1, v_2, e_6, v_4, e_7, v_3$ is a simple path from v_1 to v_3 ;
- $v_2, e_4, v_3, e_7, v_4, e_5, v_1, e_1, v_2$ is a simple circuit;
- $v_3, e_4, v_2, e_6, v_4, e_5, v_1, e_1, v_2, e_1, v_1$ is merely a walk;
- $v_4, e_5, v_1, e_1, v_2, e_4, v_3, e_2, v_2, e_6, v_4$ is a circuit which is not simple.

The walk

$$v_3, e_4, v_2, e_6, v_4, e_5, v_1, e_1, v_2, e_1, v_1$$

from v_3 to v_1 may be “extracted” from the above graph and shown separately.



Clearly, if we wish to get from v_3 to v_1 this is unnecessarily complicated, and we can replace it by a simple path

$$v_3, e_4, v_2, e_1, v_1 .$$

Theorem. Let u, v be vertices in G . There is a walk from u to v if and only if there is a simple path from u to v .

Proof. Suppose that G contains a walk from u to v , and let

$$v_0, e_1, v_1, e_2, v_2, \dots, v_{n-1}, e_n, v_n \quad (*)$$

be the shortest such walk, where we have written $v_0 = u$ and $v_n = v$.

Suppose that some vertex v_i is repeated, say $v_i = v_j$. If $i > 0$ then

$$v_0, e_1, v_1, \dots, v_{i-1}, e_i, v_j, e_{j+1}, v_{j+1}, \dots, v_n$$

is a walk from u to v ; if $i = 0$ then

$$v_j, e_{j+1}, v_{j+1}, \dots, v_n$$

is a walk from u to v . In each case the new walk is of shorter length than the previous one, which is impossible. Hence no vertex is repeated in the walk $(*)$.

Thus all of the vertices v_k are different, and so the edges e_k are all different too. Therefore $(*)$ is a simple path from u to v .

The converse is clear, since any simple path is a special type of walk.

Corollary. Let G be a graph with n vertices. If there is a walk from u to v then there is a walk of length at most $n - 1$ from u to v .

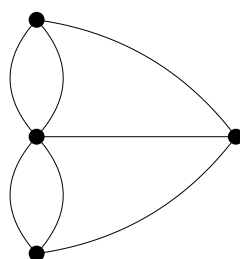
Proof. If there is a walk from u to v there is a simple path from u to v ; in such a path there are no repeated vertices, and so there can be at most n vertices; therefore there are at most $n - 1$ edges.

Definition. A graph G is said to be **connected** if there is a walk between any two distinct vertices of G .

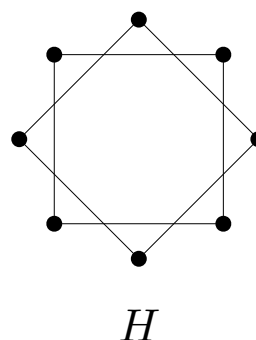
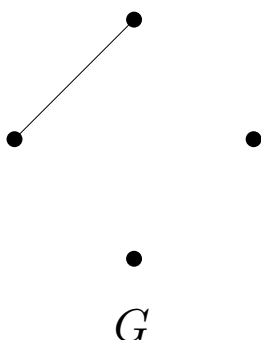
The **connected components** of a graph are its maximal connected subgraphs.

Note. From previous results, we may say that G is connected if there is a *simple path* between any two distinct vertices of G ; or if there is a walk of length at most $n - 1$ between any two such vertices, where n is the number of vertices of G .

Examples. This graph



is connected, while these are not.



Here G has three connected components; H has two. A connected graph has just one component.

Theorem. *Removing an edge from a circuit.* If G is a connected graph and an edge which is part of a circuit in G is removed, the remaining graph H is still connected.

Proof. Suppose that we remove an edge f_1 which is part of the circuit

$$f_1, f_2, \dots, f_k$$

in G . Let the endpoints of f_i be v_{i-1} and v_i , with $v_0 = v_k$. Since the circuit has no repeated edges, $f_2, \dots, f_k$ are all different from f_1 .

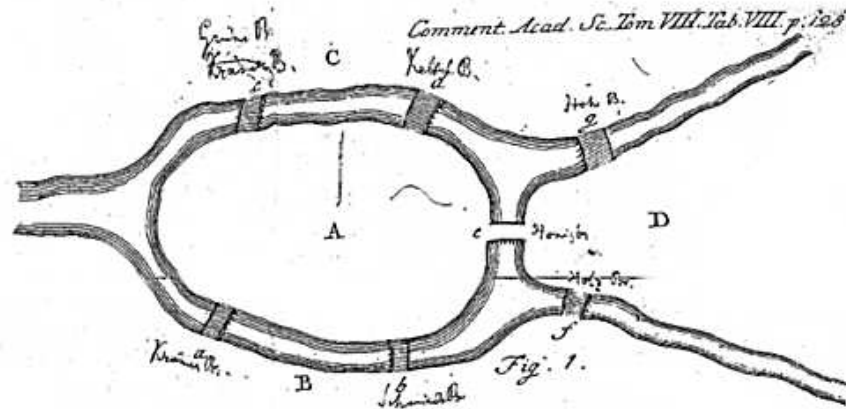
Let u, v be vertices in H . Then there is a simple path

$$e_1, e_2, \dots, e_m$$

from u to v in G . Denote the endpoints of e_i by u_{i-1} and u_i . Suppose that the edge f_1 occurs in this path, say $e_j = f_1$. Then either $u_{j-1} = v_0$, $u_j = v_1$, and we replace e_j by $f_k, \dots, f_3, f_2$; or $u_{j-1} = v_1$, $u_j = v_0$, in which case we replace e_j by $f_2, f_3, \dots, f_k$. After making these substitutions we have a path in H from u to v .

Since this holds for any vertices u, v in H , we have shown that H is a connected graph.

Question. How do we translate the Königsberg bridges problem into graph theory?



Definition. An **Euler circuit** in a graph G is a circuit containing every edge and every vertex of G .

Theorem. *Necessary condition for an Euler circuit.* If a graph contains an Euler circuit then every vertex is of even degree.

Proof. The circuit leaves its initial vertex v_0 along one edge at the beginning of the path, and arrives back along a different edge at the end. This accounts for two edges incident on v_0 . Each time the circuit visits another vertex (including the initial vertex if it is revisited), two new edges are used. Therefore the degree of each vertex is twice the number of times it is visited, which clearly is even.

Problem. Is the converse of this theorem true?

Theorem. *Existence of an Euler circuit.* A connected graph in which every vertex has even degree has an Euler circuit.

Proof. Let G be a connected graph in which every vertex has even degree. It is easy to see that the result is true if G has only one vertex, so we may assume that G has at least two vertices. Let C be a circuit with maximum possible length in G . We shall prove by contradiction that C contains every edge of G and is therefore an Euler circuit.

So, suppose that there is an edge $e = uv$ in G which is not in C . Since G is connected, there is a path $u, e, v, \dots$ which eventually reaches a vertex on the circuit C ; this path must, somewhere, reach the circuit for the first time, and so there is an edge $f = wx$ in G such that w is a vertex in C and f is not an edge in C .

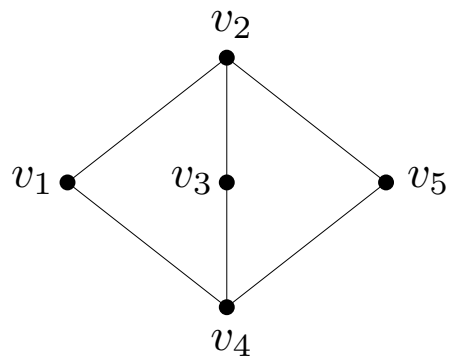
Now form a path P starting with $w, f, x, \dots$ and continuing as far as possible, using only edges which are not in C . Note that every vertex has even degree in G ; and ignoring edges already used in C will decrease the degree by an even number; so each vertex has an even number of unused edges adjacent on it, and the only vertex where P is forced to stop is w . That is, P is in fact a circuit.

Now consider the path $C \cup P$ formed as follows: start at w ; follow the circuit C until it returns to w ; then follow P until it returns to w again. Since $C \cup P$ has no repeated edges it is a circuit; and it is longer than C , which is a contradiction. Therefore C contains every edge in G .

Finally, since G is connected and has more than one vertex, it cannot have any vertices without incident edges; therefore C also contains every vertex of G , and, by definition, C is an Euler circuit. This completes the proof.

Definition. An **Euler path** from u to v in a graph G is a path from u to v which includes every edge and every vertex of G .

Example.



The above graph has an Euler path

$$v_2, v_1, v_4, v_3, v_2, v_5, v_4 .$$

(Note that since the graph is simple, we may omit the edge labels and describe the path by means of the vertices it passes through, with no risk of ambiguity.)

Theorem. *Existence of an Euler path.* Let u and v be vertices, $u \neq v$, in a connected graph G . Then there is an Euler path in G from u to v if and only if u, v have odd degrees and each other vertex of G has even degree.

Proof. Consider the graph H obtained by adding an edge e between u and v . (Or adding a parallel edge if there already is one.) In H every vertex has even degree and there is therefore an Euler circuit

$$e, e_1, \dots, e_m,$$

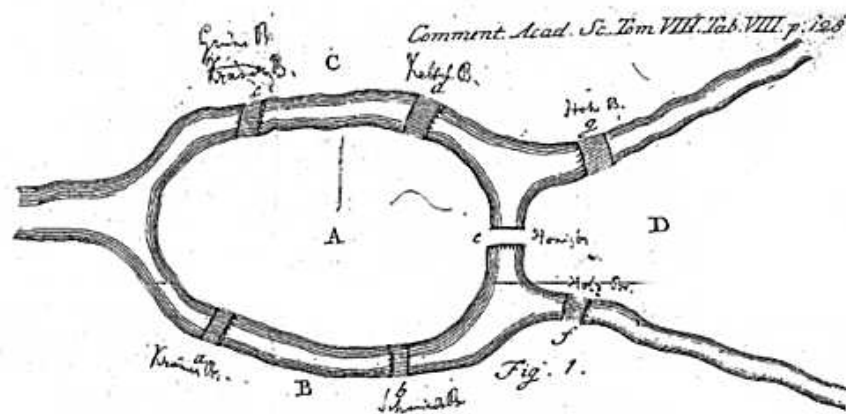
where, say, e_m is incident on u and e_1 is incident on v . Hence G contains an Euler path

$$e_m, e_{m-1}, \dots, e_1$$

from u to v .

Exercise. Complete this proof by showing that the converse is also true.

Solution of the Königsberg bridges problem. Draw a graph with a vertex for each separate land mass and an edge between two vertices for each bridge joining the corresponding regions of land. Since this graph has odd vertices, it is impossible to cross all the bridges once each in a single continuous path returning to the starting point. Since the number of odd vertices is greater than 2, it is not even possible to cross all the bridges once each in a journey with different starting and finishing points.



Historical notes. Euler stated and solved this problem before the Academy of Sciences of St. Petersburg, Russia, on August 26, 1735, and his article was published by the Academy in 1741. Although it is often asserted that this article marks the beginning of graph theory, Euler's arguments have little resemblance to modern graph theory and in fact take more of a combinatorial approach.

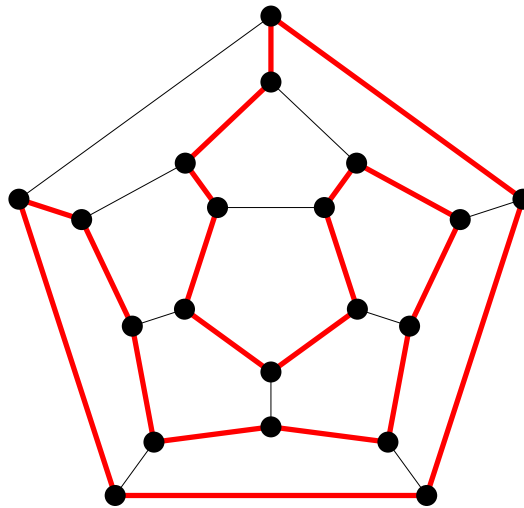
At math.dartmouth.edu/~euler/docs/originals/E053.pdf you can read Euler's original article.

See www.cs.berkeley.edu/~christos/classics/euler.pdf for an English translation together with an interesting discussion.

Definition. A **Hamilton circuit** for a graph G is a circuit which includes each vertex of G once and only once. A **Hamilton path** from u to v is a path from u to v which includes every vertex of G once and only once.

Examples.

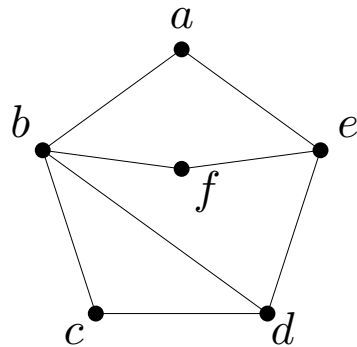
- The following graph has a Hamilton circuit, shown as a red line.



- K_n has a Hamilton circuit if...
- $K_{m,n}$ has a Hamilton circuit if...
- W_n has a Hamilton circuit if...
- Q_n has a Hamilton circuit if...

Note. There is no known simple method of determining whether or not a given graph has a Hamilton circuit or a Hamilton path!

Example. Show that the following graph has no Hamilton circuit. Does it have a Hamilton path?



Answer. Suppose that the graph has a Hamilton circuit. In order that vertex a be included, the circuit must contain edges ab and ae . Likewise it must contain bf and ef . But then it is impossible for the circuit to include vertices c and d without visiting b or e again.

The graph contains various Hamilton paths, for example $abcdef$.

Theorem (Dirac, 1952). *Sufficient condition for a Hamilton circuit.* A connected simple graph G with n vertices, $n \geq 3$, has a Hamilton circuit if each vertex has degree at least $\frac{1}{2}n$.

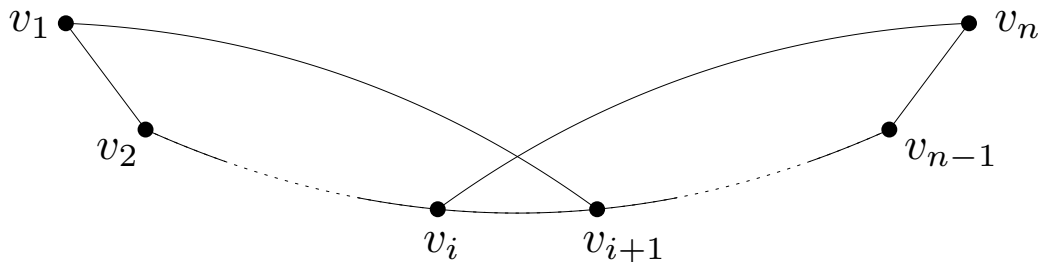
Proof by contradiction. Assume that G is a connected simple graph with $n \geq 3$ vertices which has each vertex of degree at least $\frac{1}{2}n$ but has no Hamilton circuit. By adding edges to G we obtain a graph H which still has no Hamilton circuit, but does have a Hamilton path.



Note that v_1 is not adjacent to v_n . Now v_1 is adjacent to some of the vertices $v_3, \dots, v_{n-1}$. Consider the vertices *immediately preceding* these on the Hamilton path: these will be in $\{v_2, \dots, v_{n-2}\}$. Also, consider the vertices of $\{v_2, \dots, v_{n-2}\}$ which are adjacent to v_n . Altogether, the two sets considered include

$$\deg(v_1) - 1 + \deg(v_n) - 1 \geq n - 2$$

vertices. But there are only $n - 3$ elements in $\{v_2, \dots, v_{n-2}\}$, so some vertex v_i must be in both sets, and we have a Hamilton circuit $v_1 v_{i+1} v_{i+2} \cdots v_{n-1} v_n v_i v_{i-1} \cdots v_2 v_1$ in H . But this is a



contradiction since H has no Hamilton circuit; therefore G must have had a Hamilton circuit, and the proof is complete.

Note. The converse of the preceding theorem is false.

Examples.

- A simple graph with vertices of degrees 3, 3, 3, 4, 4, 5 has a Hamilton circuit.
- A simple graph with seven vertices, each of degree 4, has a Hamilton circuit.
- If there are six towns, each joined directly by road to three or more of the others, then there is a tour starting in one town, visiting each town exactly once, and returning to the first town.

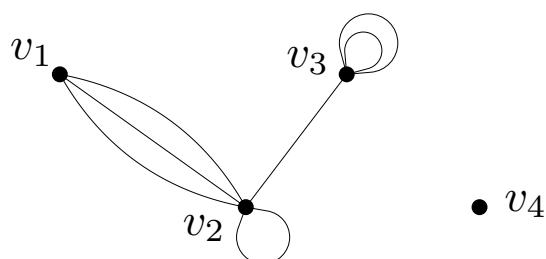
Theorem (Ore, 1960). Let G be a connected simple graph with n vertices, $n \geq 3$. If for every pair v_1 and v_2 of non-adjacent vertices $\deg(v_1) + \deg(v_2) \geq n$, then G has a Hamilton circuit.

Proof. Exercise: very similar to the previous proof.

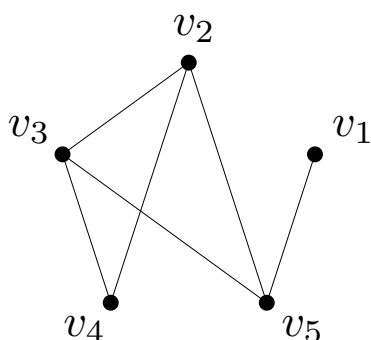
ADJACENCY MATRIX OF A GRAPH

Definition. The **adjacency matrix** of a graph G with vertices $v_1, v_2, \dots, v_n$, listed in that order, is the $n \times n$ matrix A with (i, j) element a_{ij} equal to the number of edges $v_i v_j$ in G .

Examples. The adjacency matrices of the following graphs are as shown.



$$\begin{pmatrix} 0 & 3 & 0 & 0 \\ 3 & 1 & 1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$



$$\begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \end{pmatrix}$$

Note.

- A graph may have different adjacency matrices if the vertices are listed in a different order.
- Any adjacency matrix is symmetric.
- An adjacency matrix for a simple graph consists of zeros and ones only; its main diagonal is zero throughout.

Conversely, any symmetric matrix with entries from $\mathbb{N}$ can be interpreted as the adjacency matrix of some graph.

Exercise. Draw a graph with adjacency matrix

$$\begin{pmatrix} 1 & 0 & 2 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 2 & 0 \end{pmatrix}.$$

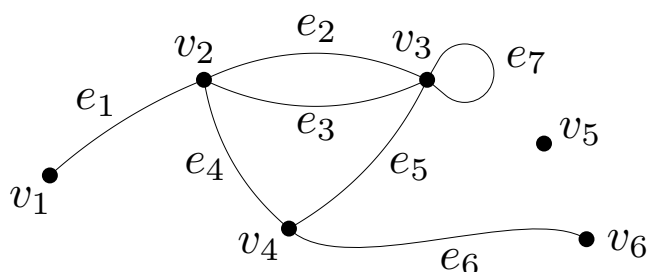
Answer.

Another way of doing all this is with an incidence matrix.

Definition. A graph G with vertices $v_1, v_2, \dots, v_n$ and edges $e_1, e_2, \dots, e_m$ has the $n \times m$ **incidence matrix** M with (i, j) element

$$m_{ij} = \begin{cases} 1 & \text{if } e_j \text{ is incident on } v_i \\ 0 & \text{otherwise.} \end{cases}$$

Example. The incidence matrix of



is

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

Note. Just by looking at the matrix, we can see that

- e_2 and e_3 are parallel edges;
- e_7 is a loop;
- v_5 is isolated.

One of the main applications of matrices to graph theory is in counting walks.

Example. In a graph with five vertices $v_1, \dots, v_5$, how can we count all walks of length 2 from v_4 to v_2 ?

Solution. A walk of the required type will have an edge from v_4 to v_k and another from v_k to v_2 , for some k . For each k , the number of possibilities is

$$\left(\begin{array}{c} \text{number of edges} \\ \text{from } v_4 \text{ to } v_k \end{array} \right) \times \left(\begin{array}{c} \text{number of edges} \\ \text{from } v_k \text{ to } v_2 \end{array} \right) ,$$

that is, $a_{4k}a_{k2}$, where A is the adjacency matrix of the graph. So the total number of walks is

$$\sum_{k=1}^5 a_{4k}a_{k2} .$$

This is the 4th row, 2nd column entry of A^2 !

Theorem. *Counting walks.* Let G be a graph with vertices $v_1, \dots, v_n$, and let A be the adjacency matrix of G . Then the number of walks of length m in G from v_i to v_j is the (i, j) element of A^m .

Proof. The result is true for $m = 1$, since $A^1 = A$ and a walk of length 1 is just a single edge.

Suppose the result is true for some particular m ; write $A^m = B$ and $A^{m+1} = C$. We have assumed

“there are b_{ij} walks of length m from v_i to v_j ”

and we wish to show

“there are c_{ij} walks of length $m + 1$ from v_i to v_j ”.

To get from v_i to v_j in $m + 1$ steps we first go from v_i along any edge to some v_k , then in m steps from v_k to v_j . For each k this gives $a_{ik}b_{kj}$ possibilities. So the total number of paths is

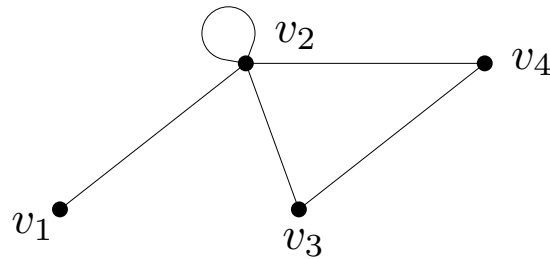
$$\sum_{k=1}^n a_{ik}b_{kj} .$$

But $A^{m+1} = A A^m$; that is, $C = AB$; so by matrix multiplication,

$$c_{ij} = \sum_{k=1}^n a_{ik}b_{kj} ,$$

as required. By induction, the result is true for all integers $m \geq 1$.

Example. The graph



has adjacency matrix

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix},$$

and calculation gives

$$A^4 = \begin{pmatrix} 4 & 9 & 6 & 6 \\ 9 & 25 & 15 & 15 \\ 6 & 15 & 10 & 9 \\ 6 & 15 & 9 & 10 \end{pmatrix}.$$

Therefore the number of ways of getting from v_3 to v_2 in four steps is 15, and the number of ways of getting from v_1 to v_3 in four steps is 6. **Check:** the six paths are in fact

$$\begin{aligned} &v_1v_2v_1v_2v_3, \quad v_1v_2v_2v_2v_3, \quad v_1v_2v_2v_4v_3, \\ &v_1v_2v_3v_2v_3, \quad v_1v_2v_3v_4v_3, \quad v_1v_2v_4v_2v_3. \end{aligned}$$

Theorem. *Adjacency matrix of a connected graph.* Let G be a graph with vertices $v_1, \dots, v_n$ and let A be its adjacency matrix. Let

$$C = I + A + A^2 + \dots + A^{n-1} .$$

Then G is connected if and only if C has no zero entries.

Proof. First, note that all entries of $I, A, \dots, A^{n-1}$ are non-negative integers.

Suppose G is connected. Then any two vertices v_i, v_j are connected by some walk of length $k \leq n - 1$. Hence the (i, j) entry of A^k is non-zero; and so the (i, j) entry of C is non-zero.

Conversely, if $c_{ij} \neq 0$ then some power A^k has a non-zero (i, j) entry, which means there is a walk (of length k) from v_i to v_j .

Example. Let G be the graph with adjacency matrix

$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 2 & 0 \\ 0 & 2 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Calculation shows that

$$I + A + A^2 + A^3 = \begin{pmatrix} 3 & 7 & 4 & 3 \\ 7 & 10 & 16 & 1 \\ 4 & 16 & 16 & 2 \\ 3 & 1 & 2 & 2 \end{pmatrix},$$

which has no zero entries. Therefore G is connected.

Exercise. Check this by drawing the graph of G .

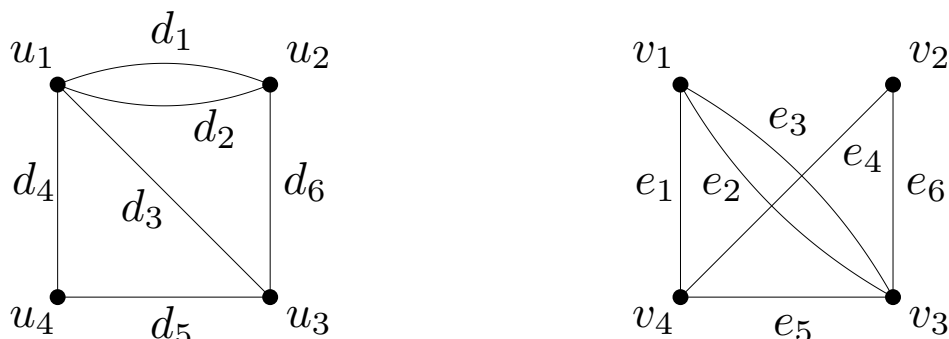
ISOMORPHISM OF GRAPHS

Definition. Two graphs G_1, G_2 are **isomorphic** if there exist bijections

$$f : V_1 \rightarrow V_2 \quad \text{and} \quad g : E_1 \rightarrow E_2$$

between their vertex sets and edge sets respectively, under which **incidence is preserved**: that is, for any $v \in V_1$ and $e \in E_1$ we have

$$e \text{ is incident on } v \quad \text{iff} \quad g(e) \text{ is incident on } f(v) .$$

Example.

The above graphs are isomorphic, as is shown by the following bijections.

u_1	v_1	d_1	e_1
u_2	v_2	d_2	e_2
u_3	v_3	d_3	e_3
u_4	v_4	d_4	e_4
		d_5	e_5
		d_6	e_6

It can be checked that these functions preserve incidence; for example

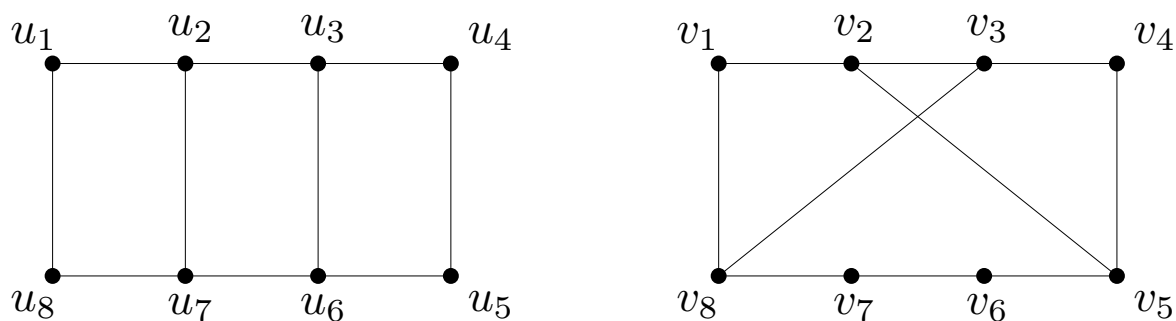
- d_1 is incident on u_1 , and $g(d_1) = e_2$ is incident on $f(u_1) = v_3$;
- d_3 is not incident on u_4 , and $g(d_3) = e_5$ is not incident on $f(u_4) = v_2$.

In the case of simple graphs the definition of isomorphism can be given in a simpler form since we don't have to worry about parallel edges or loops.

Theorem. *Isomorphism of simple graphs.* Two simple graphs G_1 and G_2 are isomorphic if and only if there exists a bijection $f : V_1 \rightarrow V_2$ which preserves adjacency, that is, u is adjacent to v in G_1 if and only if $f(u)$ is adjacent to $f(v)$ in G_2 .

Proof. Exercise.

Example. The graphs G (on the left) and H are not isomorphic.



Proof by contradiction. Suppose G and H are isomorphic. There are exactly two vertices in G adjacent to u_1 . By preservation of incidence, there are exactly two vertices in H adjacent to $f(u_1)$: that is, $f(u_1)$ has degree 2. Therefore

$$f(u_1) = v_1, v_4, v_6 \text{ or } v_7.$$

For similar reasons $f(u_8)$ must be one of the same four vertices; also, u_1 and u_8 are adjacent, so $f(u_1)$ and $f(u_8)$ are adjacent. The only possibility is that $f(u_1)$ and $f(u_8)$ are equal to v_6 and v_7 .

In the same way, since u_4 and u_5 have degree 2 in G , we must have $f(u_4)$ and $f(u_5)$ equal to v_1 and v_4 . But this is impossible because u_4 and u_5 are adjacent, while v_1 and v_4 are not.

Therefore G and H are not isomorphic.

More briefly, we can say: G has two pairs of adjacent vertices of degree 2, but H does not; so G and H are not isomorphic.

Isomorphic invariants. Since an isomorphism really only amounts to changing the names of vertices and edges, it is clear that if a graph G is (for example) connected, then so is any graph isomorphic to G . A property such as connectedness is said to be an *invariant with respect to isomorphism of graphs*. Similarly, any graph isomorphic to G has the same number of vertices as G ; thus the number of vertices of a graph is also an isomorphic invariant.

Definition. Let P be a property. Suppose that whenever G is a graph with property P and H is isomorphic to G , the graph H also has property P . Then P is called an **isomorphic invariant**.

Theorem. *Some isomorphic invariants.* The following properties (among others) are invariant with respect to isomorphism of graphs:

- (i) the number of vertices;
- (ii) the number of edges;
- (iii) the number of vertices of a given degree;
- (iv) the number of circuits of a given length;
- (v) connectedness;
- (vi) being bipartite;
- (vii) the existence of an Euler circuit;
- (viii) the existence of a Hamilton circuit.

Proof of (v). Suppose that G is a connected graph and that H is isomorphic to G . Let f, g be functions between the vertices and the edges, respectively, of G and H , which preserve incidence.

Let v, v' be vertices of H . Since f is onto, we have $v = f(u)$, $v' = f(u')$ for some u, u' in G . Since G is connected there is a walk

$$u_0, e_1, u_1, \dots, u_{m-1}, e_m, u_m$$

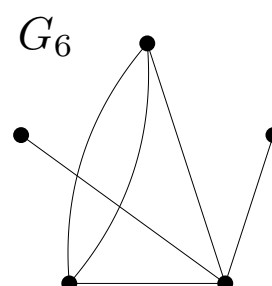
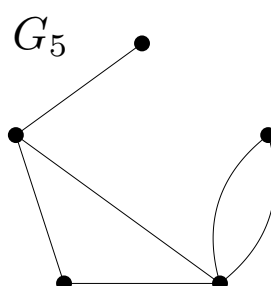
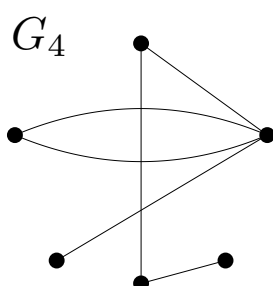
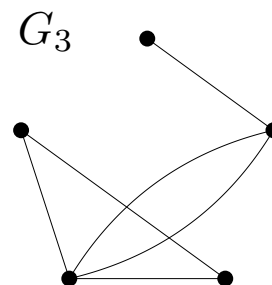
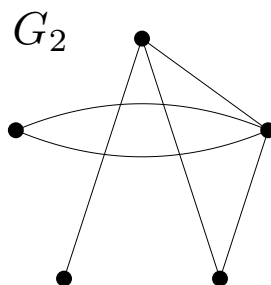
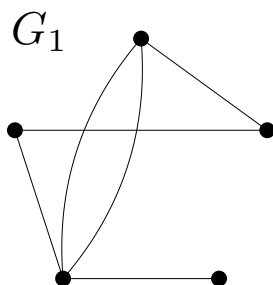
from u to u' in G . As incidence is preserved,

$$f(u_0), g(e_1), f(u_1), \dots, f(u_{m-1}), g(e_m), f(u_m)$$

is a walk from $f(u) = v$ to $f(u') = v'$ in H .

We have shown that for any vertices v, v' in H there is a walk from v to v' ; thus H is connected.

Exercise. Which, if any, of the following are isomorphic?



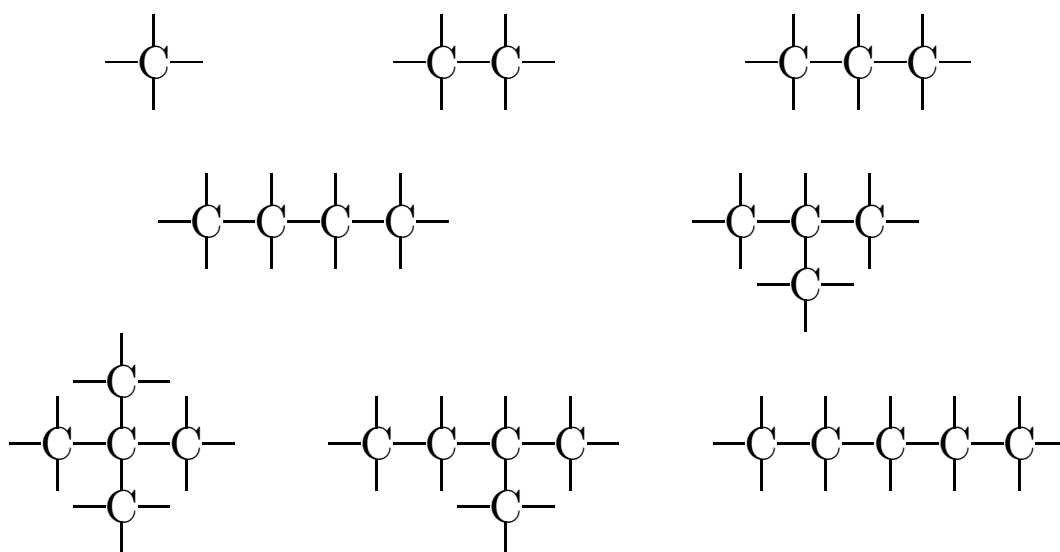
Solution.

- G_4 is not isomorphic to any of the others as it has six vertices, while they have five.
- G_6 has vertices of degrees 1, 1, 3, 3, 4, and so is not isomorphic to the others.
- G_1 has no circuit of length 3; G_2, G_3, G_5 do.
- G_2 is seen to be isomorphic to G_5 by appropriate labelling.
- G_3 passes all the tests of the previous theorem for being isomorphic to G_2 , yet is not! **Exercise.** Why not?

Note. Even if two graphs share all of the properties of the theorem, they may still fail to be isomorphic.

TREES

Saturated hydrocarbons are molecules consisting of carbon and hydrogen atoms only, in which there are as many hydrogen as possible for a given number of carbon atoms. The few smallest look like this:



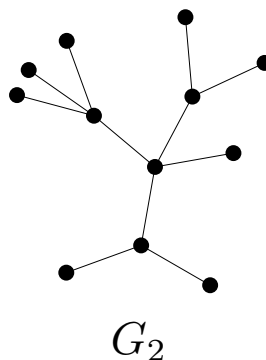
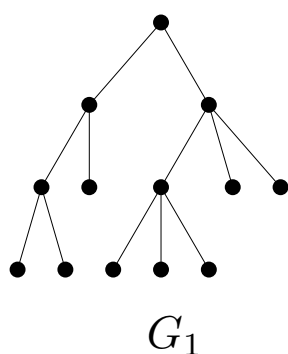
Such molecules can be represented as graphs by drawing a vertex for each carbon atom and an edge for each carbon-carbon bond. Such graphs

- are connected;
- have no simple circuits.

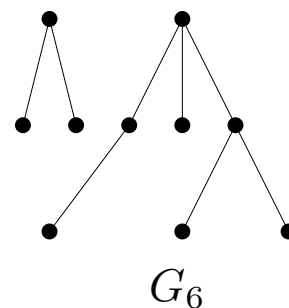
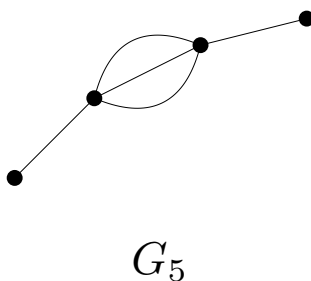
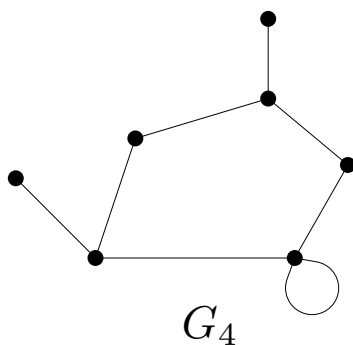
They were studied by Arthur Cayley in the 19th century in order to find out the number of saturated hydrocarbon *isomers* having a given number of carbon atoms.

Definition. A **tree** is a connected graph with no circuits of length 1 or more.

Examples. The following are trees,



while these are not...



... although G_6 is a **forest**.

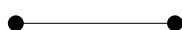
Theorem. *Trees and paths.* A graph G is a tree if and only if there exists a unique simple path between any two of its vertices.

Sketch of proof. Suppose that G is a tree; let u, v be vertices of G . Since G is connected, there exists a path between u and v , hence also a simple path. If there are two such paths we can find a circuit in G starting at the first point where the two simple paths diverge. As G is a tree this is impossible, so there must be only one simple path from u to v .

Conversely, let G be a graph in which there is a unique simple path between any two vertices. Suppose that G is not a tree. Then G must contain a simple circuit. Any two vertices on such a circuit are joined by (at least) two simple paths. This contradiction shows that G must be a tree.

Lemma. *Vertices of degree 1.* Any tree on n vertices, $n \geq 2$, has at least two vertices of degree 1.

Proof by strong induction. The result is true for $n = 2$ as there is only one possible tree on two vertices.



Suppose the result true for trees on $2, 3, \dots, n$ vertices, where n is some specific integer. Let G be a tree with $n + 1$ vertices; we must show that G has two or more vertices of degree 1.

Remove an edge uv from G ; since G has no circuits this gives two trees G_1 and G_2 (with, say, u in G_1 and v in G_2).

Case 1, G_1 has two or more vertices. Then by induction G_1 has two vertices of degree 1 in G_1 . One of these might be u , but the other has degree 1 in G .

Case 2, G_1 has only one vertex u . Then in G the only edge incident on u is uv , and so u has degree 1.

In either case G_1 contains a vertex which was of degree 1 in G . Similarly, so does G_2 . Therefore G has two (or more) vertices of degree 1, and this completes the induction step.

Theorem. *Number of edges in a tree.* For each $n \geq 1$, any tree with n vertices has $n - 1$ edges.

Proof by induction. A tree with one vertex has no edges, so the result is true for $n = 1$.

Assume the result true for some particular n . Let G be a tree with $n + 1$ vertices: we must show that G has n edges. So let v be a vertex of degree 1 in G ; remove v and the edge incident on it. What remains is a tree on n vertices, which by the inductive assumption has $n - 1$ edges. Restoring the edge we deleted earlier, G clearly has n edges.

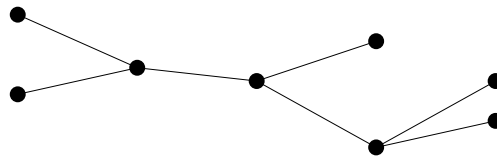
By induction, the result is true for all $n \geq 1$.

Examples.

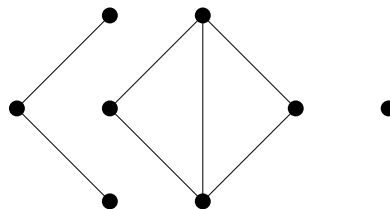
- (i) Is a graph with 8 vertices and 11 edges a tree?
- (ii) Is a graph with 8 vertices and 7 edges a tree?

Answers.

- (i) No, by the previous theorem.
- (ii) Maybe...



... but not necessarily.



Note. The converse of the above theorem is not true...

...though it is in fact true for *connected* graphs.

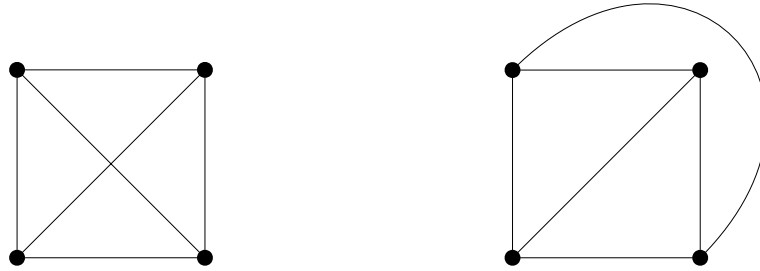
Theorem. Let G be a connected graph with n vertices. Then G is a tree if and only if it has $n - 1$ edges.

Example. Find (up to isomorphism) all trees having six vertices.

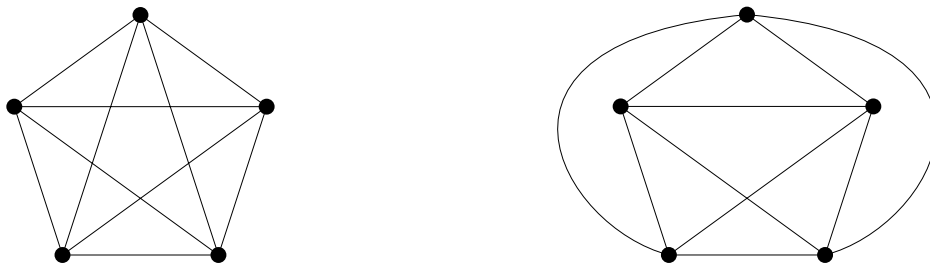
Solution. A tree on six vertices has five edges, and therefore total degree 10. Since any tree has at least two vertices of degree 1, the possible degrees, and the corresponding trees, are...

PLANAR GRAPHS

Recall that in the graph K_4 the edges do not intersect at the “central” point; this can be seen by redrawing the graph as at right.

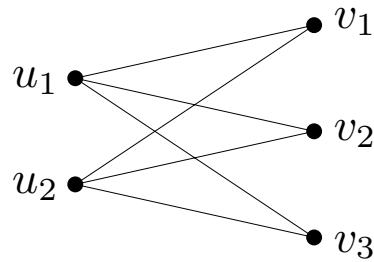


Similarly, K_5 can be redrawn in order to eliminate some of the unwanted crossings, but not all of them.



Definition. A graph G is said to be **planar** if it can be drawn in a plane in such a way that two edges never intersect (except at their common vertices). Such a drawing is called a **planar map** or **planar representation** of G .

Example. The following is not a planar representation of the graph $K_{2,3}$ since the edges intersect at places other than vertices.



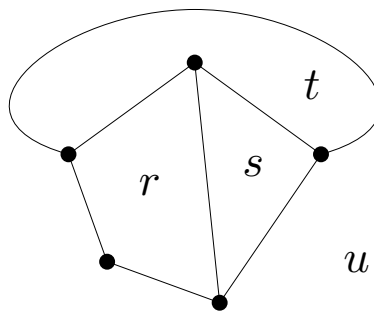
Nevertheless, $K_{2,3}$ is a planar graph since it can be redrawn isomorphically without any extraneous crossings.

The edges of a planar map unambiguously separate the plane on which it is drawn into **regions**, and we may introduce the following terminology.

Definition. The **degree of a region** in a planar map is the number of edges bounding the region, with an edge counted twice if both “sides” of it are part of the same region.

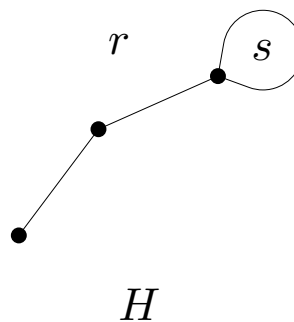
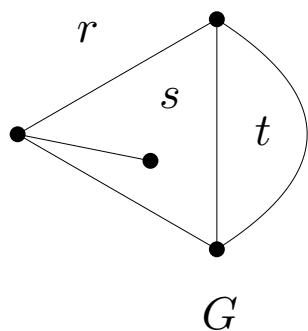
Given a planar representation of a graph G , its **dual** is the graph which has a vertex corresponding to each region of G , and one edge joining a pair of vertices for each edge of G whose “sides” are the corresponding regions.

Example. Give the degree of each region in the following map; draw the dual on the same diagram; and redraw the dual on a separate diagram.

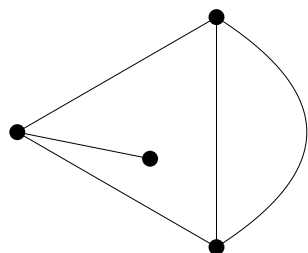


Answer. The regions r, s, t, u have degrees 4, 3, 3, 4 respectively.

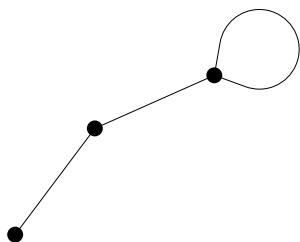
Examples. Repeat the previous questions for each of the following graphs.



Solution. In G the regions r, s, t have degrees 3, 5, 2 respectively.



In H , the region r has degree 5 and s has degree 1.



Comments.

- The degree of a region can also be defined as the length of the shortest closed walk including every edge bounding the region (with repetitions if necessary).
- The dual of a planar map is also planar, and it has the same number of edges as the original.
- By considering the dual we can see that the sum of the degrees of all regions in a planar map equals twice the number of edges.

Euler's formula. You have probably met Euler's formula for polyhedra,

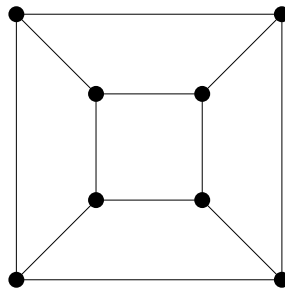
$$f + v = e + 2 .$$

Actually this is perhaps not always true – it depends on exactly what you are prepared to accept as a polyhedron. For a brilliantly argued consideration of the issues and their wider relevance to mathematical proof,

I. Lakatos, *Proofs and refutations*

is recommended reading.

Assume that a polyhedron can have one face removed and the remainder flattened onto a plane, and that the result is a *connected* planar map. For example, a cube yields the map shown.



Under these assumptions we shall show that Euler's formula holds by proving...

Theorem. *Euler's formula for planar maps.* For a non-empty connected planar map G with r regions, v vertices and e edges,

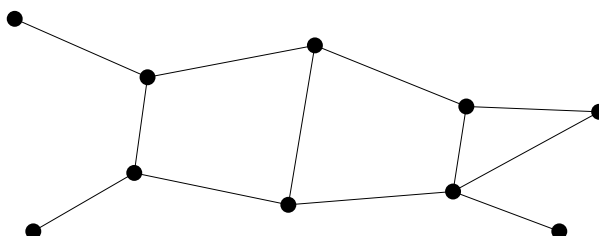
$$r + v = e + 2 .$$

Proof by induction on e . If $e = 0$ the map has just one isolated vertex; so $r = 1$, $v = 1$ and the formula holds.

Assume the formula is true for all connected planar maps with some particular number of edges $e - 1$, and for any r and v . We must prove that if G is a connected planar map with r regions, v vertices and e edges then $r + v = e + 2$.

Case 1, G has no circuits. Then $r = 1$, and since G is a tree, $e = v - 1$. Hence $r + v = e + 2$.

Case 2, G has a circuit. Remove one edge of the circuit.



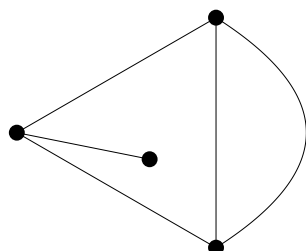
The remaining graph is still connected; and two regions have been combined into one, so there are $r - 1$ regions, v vertices and $e - 1$ edges. By the inductive assumption,

$$(r - 1) + v = (e - 1) + 2 ,$$

and so $r + v = e + 2$.

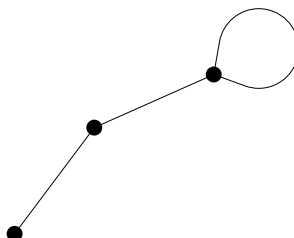
This completes the proof by induction.

Examples. We can confirm that Euler's formula holds for some previous examples.



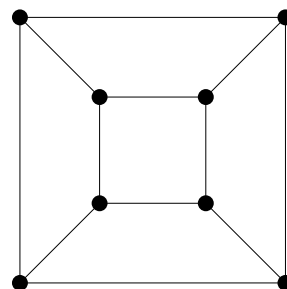
$$r = 3, v = 4$$

$$e = 5$$



$$r = 2, v = 3$$

$$e = 3$$



$$r = 6, v = 8$$

$$e = 12$$

Exercise. A connected planar map has vertices of degrees 5, 3, 3, 2, 1. Find the number of regions in the map, and draw an example of such a map.

Theorem. *Inequalities for planar graphs.* Let G be a simple connected planar graph with v vertices and e edges. Then

$$e \leq 3v - 6 .$$

Also, if G has no circuits of length 3, then

$$e \leq 2v - 4 .$$

Proof. If G is simple then no region has degree 1 or 2; hence the sum of the degrees of all the regions in a planar representation of G is at least $3r$. But we know that this sum equals twice the number of edges. Thus

$$2e \geq 3r$$

and we have

$$3(e + 2) = 3(r + v) \leq 2e + 3v$$

which simplifies to $e \leq 3v - 6$.

In case G has no circuits of length 3, each region has degree 4 or more, and so $2e \geq 4r$. The proof from this point on is similar, and is left as an exercise.

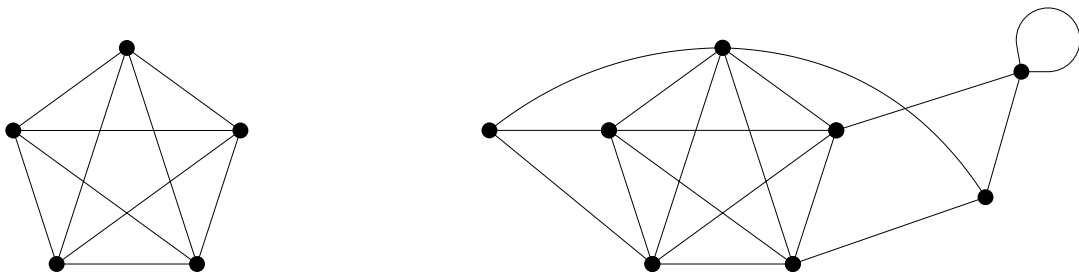
Application of Euler's formula: K_5 and $K_{3,3}$ are not planar.

Proof. In K_5 there are $v = 5$ vertices and $e = 10$ edges. Hence the inequality $e \leq 3v - 6$ is not satisfied, and K_5 is not planar.

For $K_{3,3}$ we have $v = 6$ and $e = 9$. Since $K_{3,3}$ is bipartite it contains no triangles; therefore, since the inequality $e \leq 2v - 4$ does not hold, the graph is not planar.

Exercise. Use similar methods to show that Q_n is not planar if $n \geq 4$.

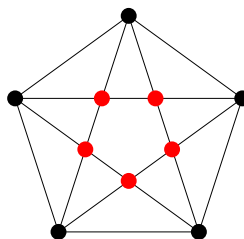
Observe that adding extra edges to a non-planar graph cannot make it planar. For example, since we know that K_5 is not planar, neither is the graph shown on the right.



Likewise, inserting vertices **of degree 2** on existing edges of a non-planar graph will not make it planar: thus the graph at right below is not planar.



Graphs related in this way are said to be **homeomorphic**. The next graph, however, is not homeomorphic to the previous ones – we have added vertices of degree more than 2 – and is planar.



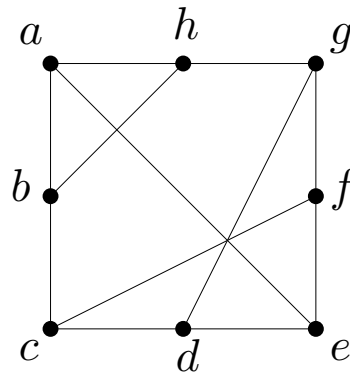
We see that any graph obtained from K_5 or $K_{3,3}$ by the given operations will be non-planar. Amazingly, the converse is also true!! – this means that, in a sense, K_5 and $K_{3,3}$ are the *only* basic non-planar graphs.

Kuratowski's Theorem. A graph is planar if and only if it does not contain as a subgraph

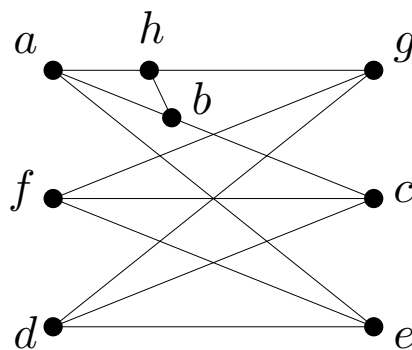
- K_5 ; or
- $K_{3,3}$; or
- any graph homeomorphic to K_5 or $K_{3,3}$.

Proof. The “only if” statement follows from the remarks just made. (Exercise: write out a more detailed proof of this.) The “if” statement is, regrettably, too hard to prove in this course.

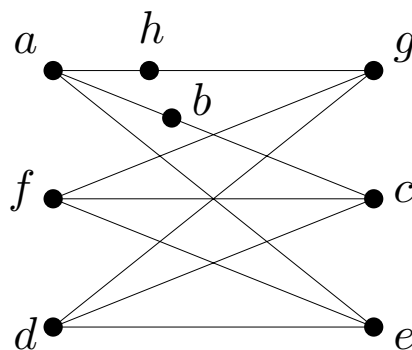
Example. Use Kuratowski's Theorem to show that the following graph is not planar.



Solution. After a lot of trial and error we redraw the graph isomorphically thus:



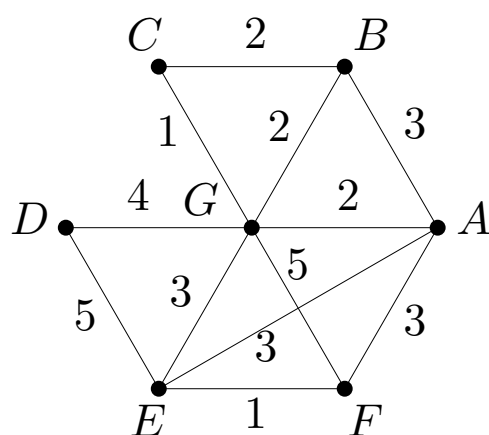
This has a subgraph



which is homeomorphic to $K_{3,3}$ since the “inserted” vertices b and h are of degree 2. By Kuratowski's Theorem, the original graph is not planar.

MINIMISATION PROBLEMS

Example. It is desired to connect up seven towns on an electric power grid. The cost (in $\$10^7$) of joining various pairs of towns is given in the accompanying diagram. Find how to build the grid with minimum cost.



Note.

- The absence of an edge between two vertices in the above graph means that the corresponding direct connection is impossible.
- The problem *does not* ask for the cheapest path from any particular town to any other town.
- There will be no solution if the given graph is not connected.
- Any grid joining the towns will be represented by a connected graph including all vertices of the given graph.
- A *minimal cost* grid will contain no circuits.
- Therefore, **any minimal cost design will be a tree containing every vertex of the above graph.**

Definition. A **weighted graph** is a graph G in which each edge e has a real number $w(e)$ called the **weight** of the edge associated with it.

A **spanning tree** for a graph G is a subgraph of G which is a tree and contains every vertex of G . A **minimal spanning tree** in a weighted graph is a spanning tree for the graph which has the smallest possible sum of weights of its edges.

It turns out that given a weighted graph G , there is a very simple algorithm for finding a minimal spanning tree in G .

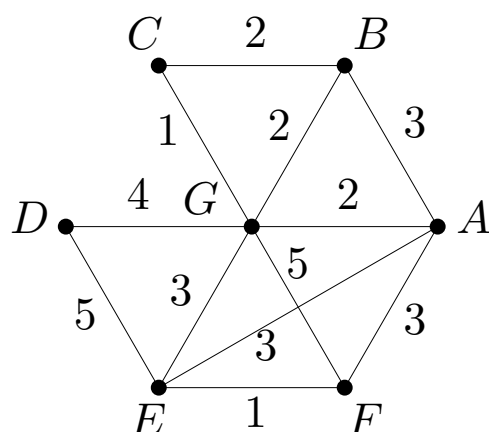
Kruskal's algorithm. Given a connected weighted graph G , the following procedure will produce a minimal spanning tree for G .

1. Let T be the graph with the same vertices as G but with no edges.
2. If T is a connected graph, the procedure is finished; otherwise, select any edge e having smallest possible weight in G .
3. If adding the edge e to T **does not** create a circuit in T , add it.
4. Remove e from G .
5. Go to step 2.

Note.

- This is sometimes called a “greedy” algorithm because at every step it takes whatever seems to be the best option available!! Such algorithms do not always give the best overall result; however this one does.
- It is necessary to prove that the algorithm actually can be executed, will terminate in a finite number of steps, and gives the correct output.
- At some points there may be more than one choice of e available. Thus there may be different possible minimal spanning trees. However the (minimal) total weight of all such trees will be the same.

Example. Solve the above electricity grid problem by using Kruskal's algorithm.



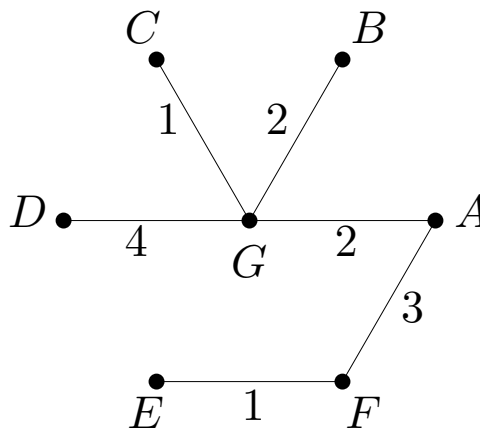
Solution. In order to find a minimal spanning tree it is convenient to draw up a table and draw T step by step.

step	edge	weight	used?
1	CG	1	yes
2	EF	1	yes
3	AG	2	yes
4	BC	2	yes
5	BG	2	no
6	AB	3	no
7	AE	3	yes
8	AF	3	no
9	EG	3	no
10	DG	4	yes

We now stop and return a minimal spanning tree. The minimal total cost for the project is $\$13 \times 10^7$.

Note.

- As mentioned previously, a minimal spanning tree need not be unique: here is another:



Observe that the total (minimal) weight is the same.

- The above trees *do not* necessarily give the cheapest connections between specific vertices. For example the cheapest path from D to E is obviously the direct one (cost 5), whereas the cost accrued along the path in the minimal spanning tree above is $4 + 2 + 3 + 1 = 10$, and in the previous one $4 + 2 + 3 = 9$.

The **shortest path problem** in a weighted graph is to find a shortest path from one given vertex to another.

Since it is not much harder we shall solve the more general problem of finding shortest paths from a given vertex to *all* other vertices. The solution to this problem turns out to be a tree again.

Definition. Let G be a weighted graph. The **weight of a path** in G is the sum of the weights of the edges in the path. The **distance** $d(u, v)$ between two vertices u, v in G is the minimum weight of any path from u to v .

If v_0 is a vertex in G , a **shortest v_0 -path spanning tree** is a spanning tree for G with the following property:

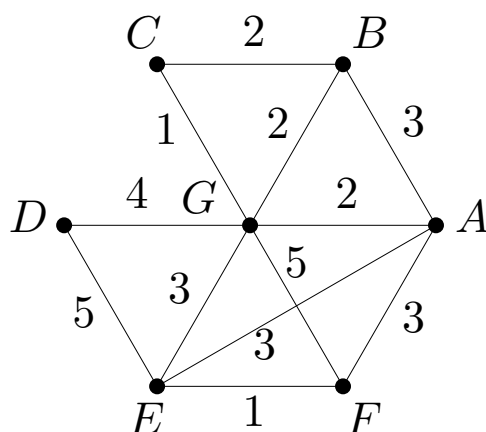
for every vertex v in G , the path in the tree from v_0 to v has no greater weight than any other path in G from v_0 to v .

Dijkstra's algorithm. Given a connected weighted graph G and a vertex v_0 in G , the following procedure will produce a shortest v_0 -path spanning tree for G .

1. Let T be the graph with vertex v_0 only and no edges.
2. If T contains all vertices of G , the procedure is finished; otherwise, go to step 3.
3. Consider all edges $e = uv$, where u is in T and v is not; choose one which gives a shortest path from v_0 through u to v ; add e and v to T .
4. Go to step 2.

Example. A freight company is based at F and wishes to serve towns A, B, C, D, E, G using the roads below (whose lengths are shown). Find the shortest routes from F to each other town.

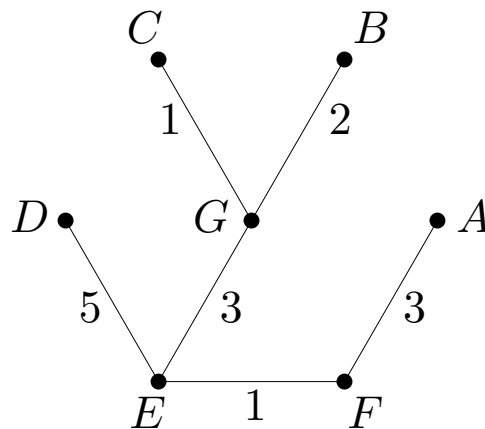
Solution. It is convenient to keep a table listing the available edges, the eventual choices and the distances from F , and to draw each edge of the graph as it is chosen.



step	available edges (total path weights)	choice	distances from F
1	FA (3), FE (1), FG (5)	FE	$d(F, E) = 1$
2	EA (4), ED (6), EG (4), FA (3), FG (5)	FA	$d(F, A) = 3$
3	AB (6), AG (5), ED (6), EG (4), FG (5)	EG	$d(F, G) = 4$
4	AB (6), ED (6), GB (6), GC (5), GD (8)	GC	$d(F, C) = 5$
5	AB (6), CB (7), ED (6), GB (6), GD (8)	AB	$d(F, B) = 6$
6	ED (6), GD (8)	ED	$d(F, D) = 6$

Note.

- We could have obtained a different result by choosing GB instead of AB at step 5.



- A shortest F -path spanning tree gives shortest paths *from F to other vertices only!* For example the distance in the tree on the previous page from B to C is 11, which is obviously not the shortest possible.
- The tree on the previous page is not a minimal spanning tree for G , since its total weight is 16, while the minimum is 13.